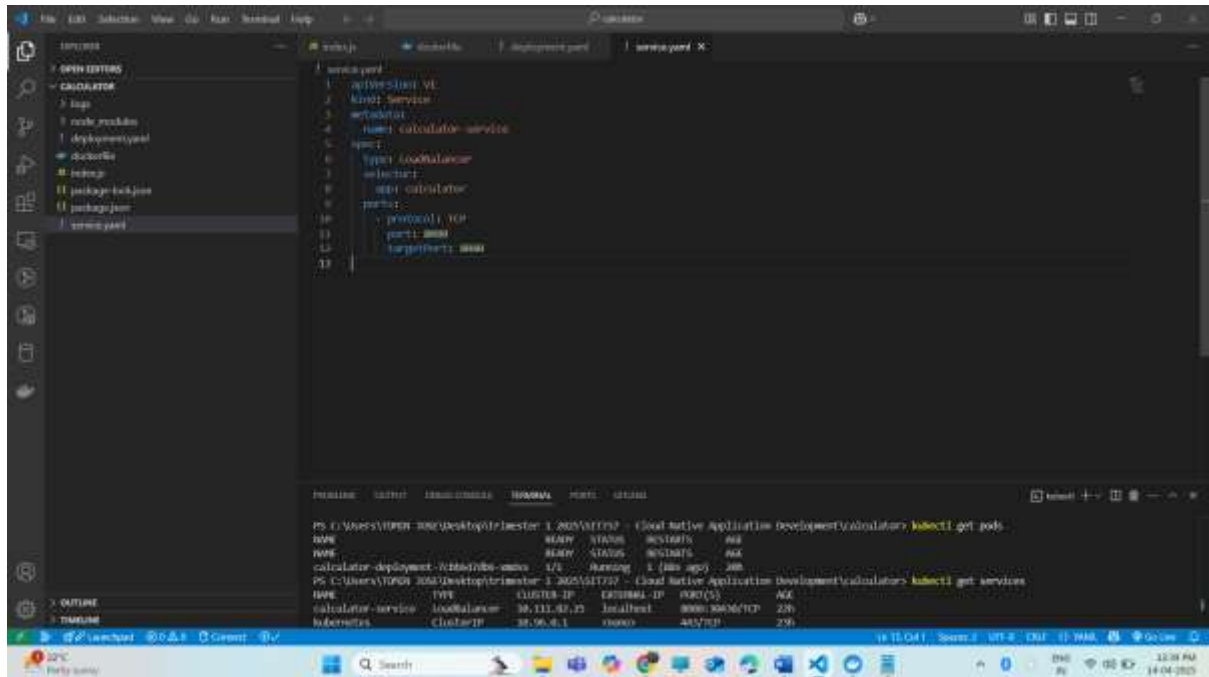


Git: <https://github.com/TOMINJOSE88/sit737-2025-prac6c.git>

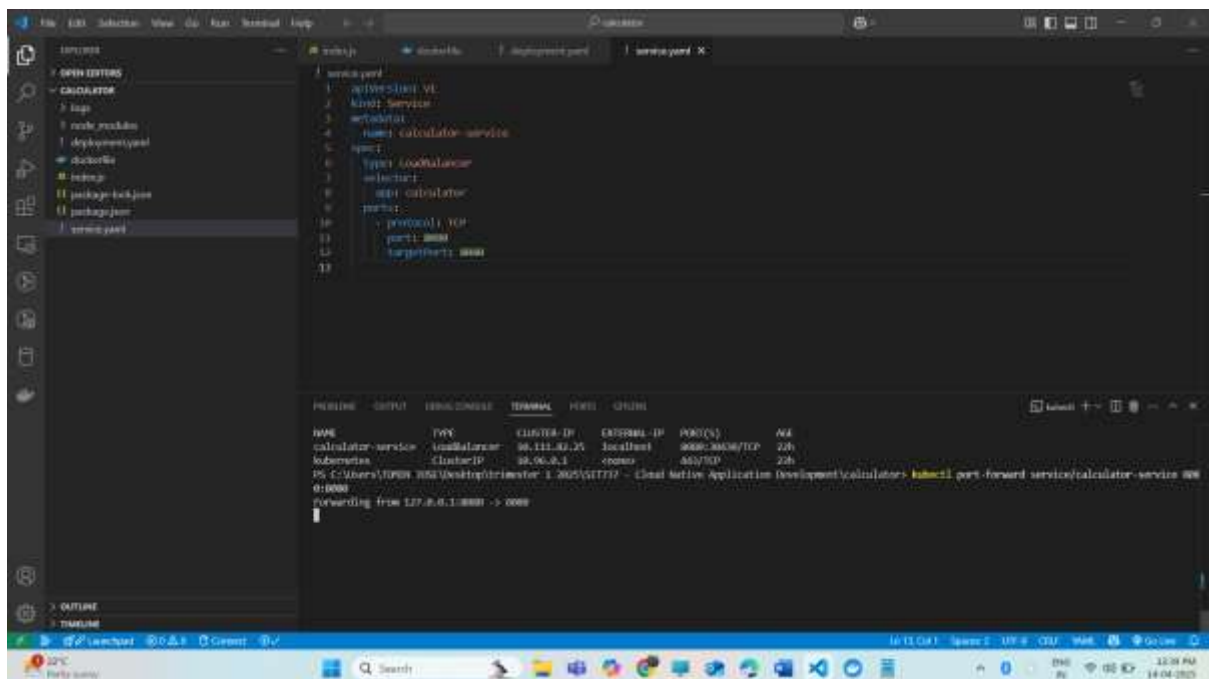
Part I

Interact with the Deployed Application:

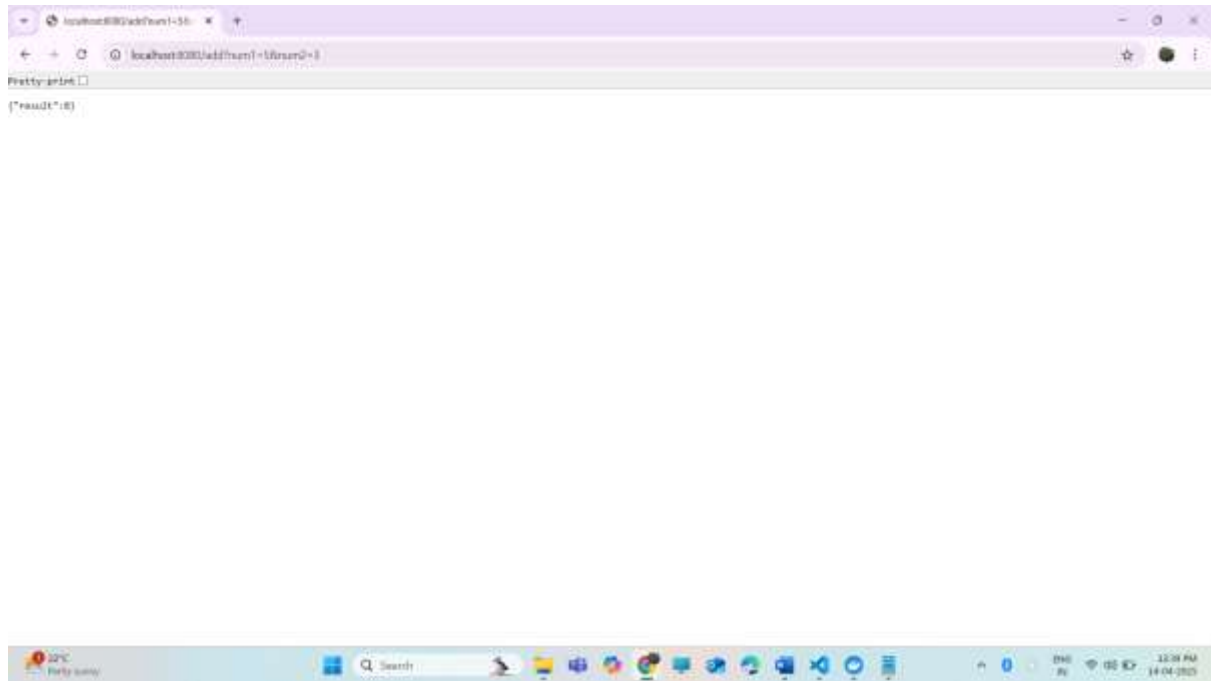
- Verify that the application is running by running `kubectl get pods` and `kubectl get services`



- Use the `kubectl port-forward` command to forward traffic from a local port to the Kubernetes service



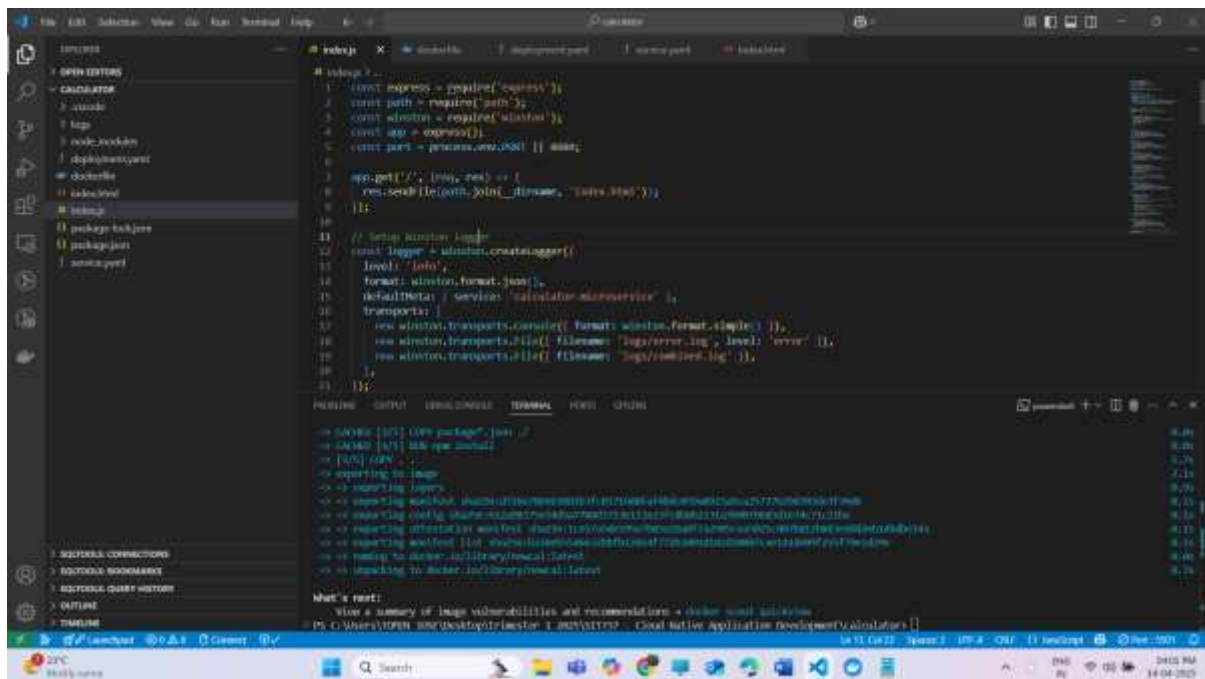
- Access the application in a web browser by navigating to localhost:



Part II

Update the application: Updating the application involves modifying the code for the Node.js application, building a new Docker image with a new version tag, and updating the Kubernetes deployment configuration to use the new image tag.

Updated code:



```
# index.js ...
1 const express = require('express');
2 const path = require('path');
3 const minston = require('minston');
4 const app = express();
5 const port = process.env.PORT || 3000;

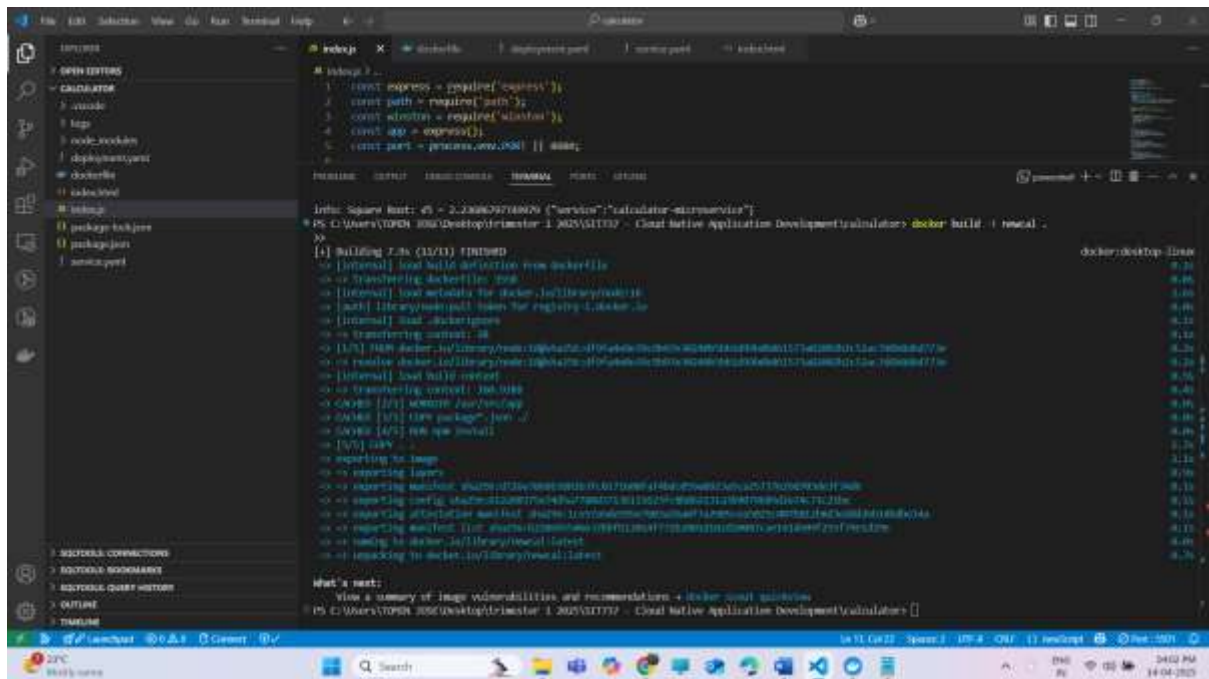
6
7 app.get('/', (req, res) => {
8   res.send({ path: path.join(__dirname, 'index.html') });
9 });

10
11 // Setup Winston logger
12 const logger = winston.createLogger({
13   level: 'info',
14   format: winston.format.json(),
15   defaultMeta: { service: 'calculator-microservice' },
16   transports: [
17     new winston.transports.Console({ format: winston.format.simple() }),
18     new winston.transports.File({ filename: 'logs/error.log', level: 'error' }),
19     new winston.transports.File({ filename: 'logs/combined.log' }),
20   ],
21 });

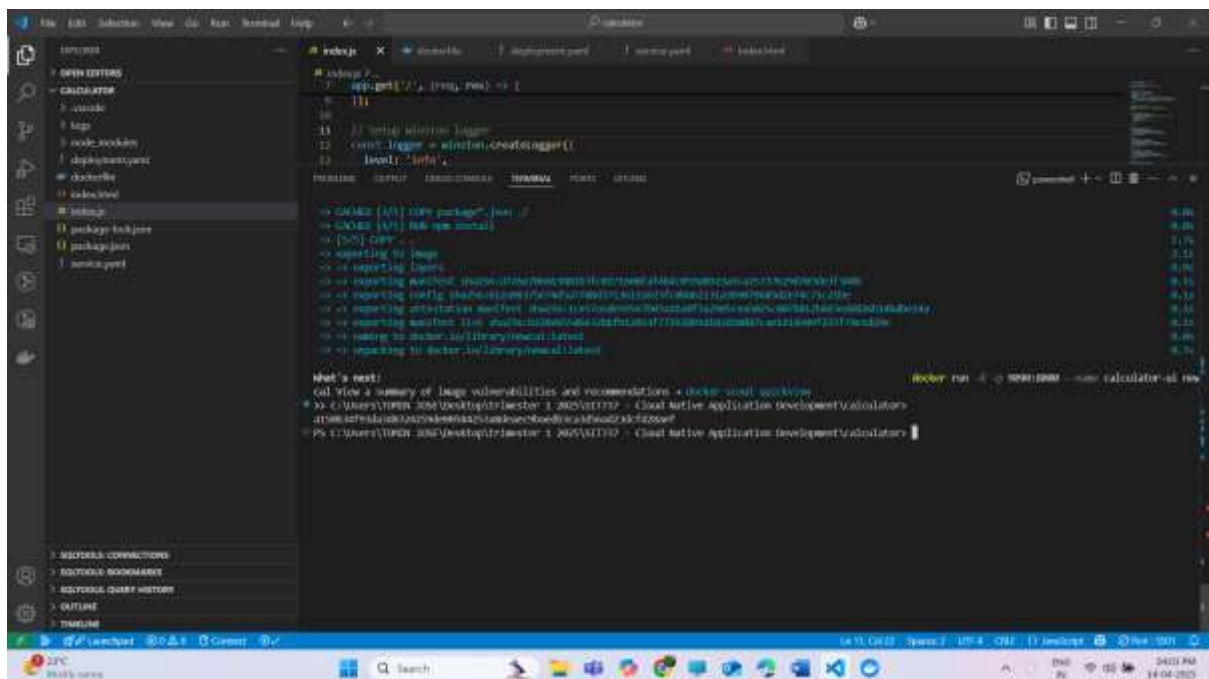
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```



Docker image:

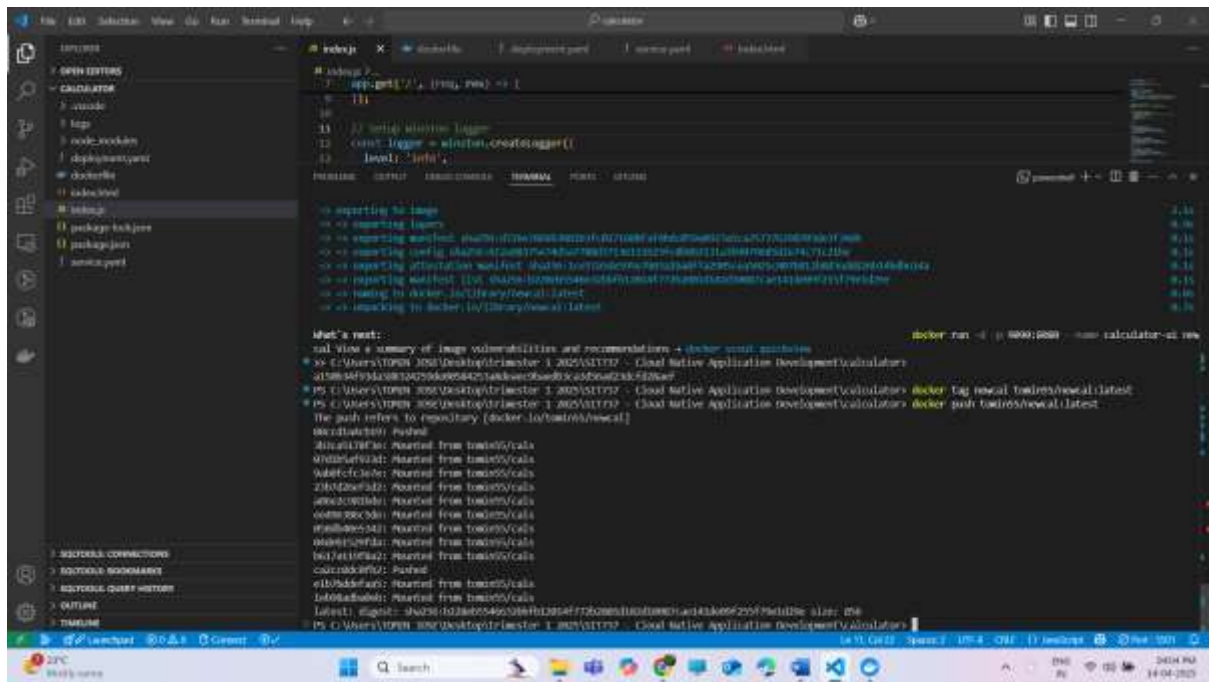


Docker Run:

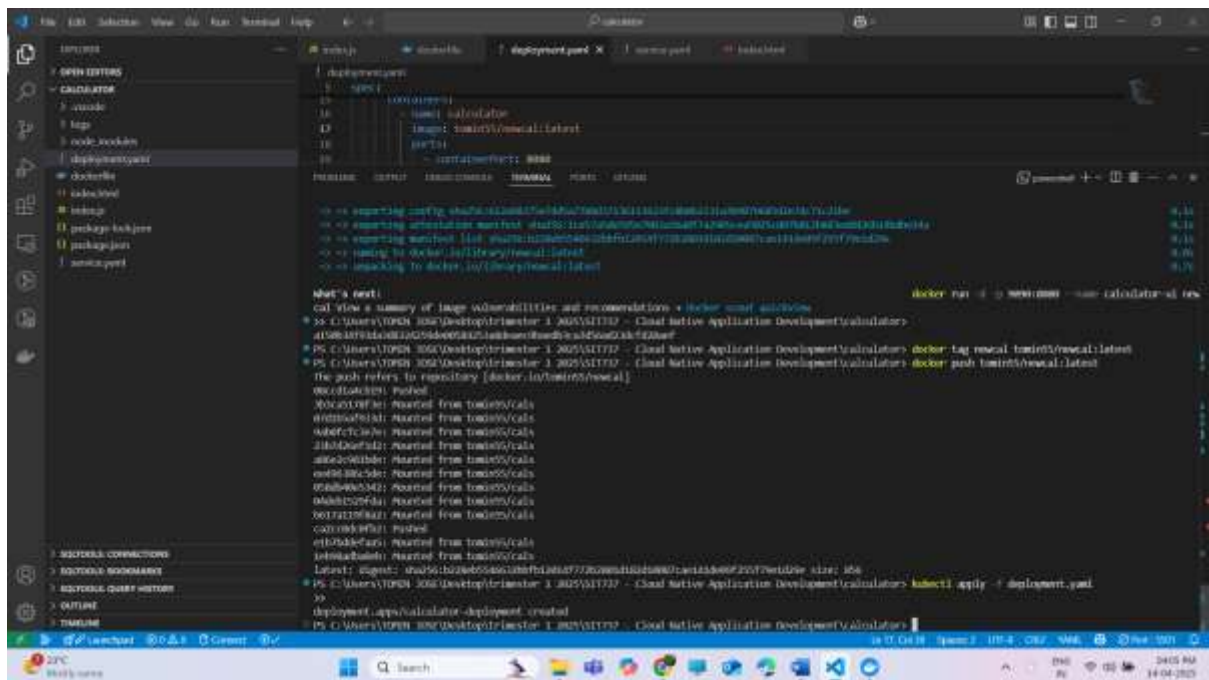




Docker Push:



Kubernetes Deployment:



The screenshot shows a VS Code editor with a file explorer on the left containing a directory structure for a calculator application. The main editor displays a `deployment.yaml` file. The terminal at the bottom shows the execution of `kubectl apply -f deployment.yaml`, which successfully creates the deployment. The output includes details about the deployment's creation and the pods being scheduled.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: calculator
spec:
  replicas: 1
  selector:
    matchLabels:
      app: calculator
  template:
    metadata:
      labels:
        app: calculator
    spec:
      containers:
      - name: calculator
        image: tomriddle/calculator:latest
        ports:
        - containerPort: 8080
```

```
PS C:\Users\TOPH> kubectl apply -f deployment.yaml
deployment.apps/calculator-deployment created
PS C:\Users\TOPH>
```

